Proof for Theorem 5 14 (Corrected Statement And Replacement Proof In The V5 Corrigendum, Appendix D; Original Label Thm:light) Use The Corrected Condition (18) With Norm(y) >= 3 Norm(x), Not The Superseded Printed Version

Generated by Statopia Prover

October 29, 2026

Statement provenance. Additional conditions were adopted automatically without human confirmation. This report describes the proof under the recorded conditions.

Key Lemmas

Lemma proved by a separate task

Lemma 1 (External Lemma 5.17: uniform HMC acceptance decay)**.** *Let d be a nonnegative integer and let $\mathcal{X}_d = \mathbb{R}^d$ be equipped with the Euclidean norm $\|\cdot\|$. Let $U : \mathcal{X}_d \to \mathbb{R}$, let $\varepsilon > 0$ be real, and let $L \geq 1$ be an integer. Let ∇U denote the Euclidean vector of partial derivatives of U. Assume:*

(i) $U(x) \geq 0$ for every $x \in \mathcal{X}_d$;

(ii) U is differentiable on $\mathcal{X}_d$;

(iii) for every real M there exists a real R such that, for every $x \in \mathcal{X}_d$,

$$\|x\| \geq R \quad \Longrightarrow \quad M \leq \frac{\|\nabla U(x)\|}{\|x\|};$$

(iv) there exists a real C such that, for all $x, y \in \mathcal{X}_d$,

$$C \le 3\|x\| \;\; \textit{and} \;\; 3\|x\| \le \|y\| \quad \Longrightarrow \quad 3\|\nabla U(x)\| \le \|\nabla U(y)\|;$$

(v) for every real B there exists a real R such that, for all $x, y \in \mathcal{X}_d$,

$$\|x\| \ge R \;\; \textit{and} \;\; 2^L\|x\| \le \|y\| \quad \Longrightarrow \quad U(x) - U(y) - \frac{1}{2}\|x\|^2 \le B.$$

For $x_0, p_0 \in \mathcal{X}_d$, define $x_{L\varepsilon}(x_0, p_0)$ and $p_{L\varepsilon}(x_0, p_0)$ by the L-step leapfrog recursion with step size ε:

$$x^{(0)} = x_0, \quad p^{(0)} = p_0, \qquad p^{(i+\frac{1}{2})} = p^{(i)} - \frac{\varepsilon}{2}\nabla U(x^{(i)}),$$

$$x^{(i+1)} = x^{(i)} + \varepsilon p^{(i+\frac{1}{2})}, \qquad p^{(i+1)} = p^{(i+\frac{1}{2})} - \frac{\varepsilon}{2}\nabla U(x^{(i+1)})$$

for $i = 0, \ldots, L-1$, and set $x_{L\varepsilon}(x_0, p_0) = x^{(L)}$, $p_{L\varepsilon}(x_0, p_0) = p^{(L)}$. Let

$$\alpha_{U,\varepsilon,L}(x_0, p_0) = \min\left\{1, \exp\left(U(x_0) - U(x_{L\varepsilon}(x_0, p_0)) + \frac{1}{2}\|p_0\|^2 - \frac{1}{2}\|p_{L\varepsilon}(x_0, p_0)\|^2\right)\right\}.$$

This is the HMC acceptance probability with mass matrix I. Then for every real $\delta < 1$, both of the following hold:

$$\lim_{\|x_0\| \to \infty} \sup_{\substack{p_0 \in \mathcal{X}_d \\ \|p_0\| \le \|x_0\|^\delta}} \alpha_{U,\varepsilon,L}(x_0, p_0) = 0,$$

and for every real $\eta > 0$ there exists a real R such that, for every $x_0 \in \mathcal{X}_d$ with $\|x_0\| \ge R$ and every $p_0 \in \mathcal{X}_d$ with $\|p_0\| \le \|x_0\|^\delta$,

$$\alpha_{U,\varepsilon,L}(x_0, p_0) \le \eta.$$

Lemma proved by task paper_53e951f48c854f4db91e6ecda697ee2b (job job_paper_53e951f48c854f4db91e6ecda697ee2b), interface revision 1. See that task's own proof and summary. Its remaining dependencies still apply; the consumer's sealed axiom interface is unchanged.

Source The cited external result is the one labelled Lemma 5.17 in the article source supplied as supp. The supplied reference does not provide an author, a more specific theorem number, or bibliographic details beyond that source label.

Reference provenance The Euclidean norm \(\|\cdot\|\) is the \(\ell_2\) norm on \(\mathbb R^d\). The mass matrix is fixed to \(M=I\), so \(p_0\sim N(0,I)\) and the kinetic energy is \(\frac12\|p\|^2\). The leapfrog recursion is \[p_{t+\varepsilon/2}=p_t-\frac{\varepsilon}{2}\nabla U(x_t),\qquad x_{t+\varepsilon}=x_t+\varepsilon p_{t+\varepsilon/2},\qquad p_{t+\varepsilon}=p_{t+\varepsilon/2}-\frac{\varepsilon}{2}\nabla U(x_{t+\varepsilon}), \] with approximate flow map \[\varphi_{L\varepsilon}(x_0,p_0):=(x_{L\varepsilon},p_{L\varepsilon}). \] The catalogue also supplies the equivalent candidate-position formula \[x_{L\varepsilon} = x_0-\frac{L\varepsilon^2}{2}\nabla U(x_0) -\varepsilon^2\sum_{i=1}^{L-1}(L-i)\nabla U(x_{i\varepsilon}) +L\varepsilon p_0, \tag{9} \] and final momentum \[p_{L\varepsilon} = p_0-\frac{\varepsilon}{2}\nabla U(x_0) -\varepsilon\sum_{i=1}^{L-1}\nabla U(x_{i\varepsilon}) -\frac{\varepsilon}{2}\nabla U(x_{L\varepsilon}). \tag{11} \] Condition (18) is the corrected condition with the factor \(3\), not the superseded printed version with \(2\). Condition (19) is a joint two-variable limit over the region \(\|y\|\ge2^L\|x\|\), with fixed \(L\); the source states no uniformity in \(L\). There is a source caveat on the \(\delta\)-range: the displayed lemma is stated for \(\delta<1\), while the replacement proof later uses the Gaussian tail conclusion \(\mu_G(\{\|p_0\|>\|x_0\|^\delta\})\to0\), which needs a positive lower bound on \(\delta\). The supplied material does not state that lower bound, so the exact admissible range of \(\delta\) is not fully resolved in the source excerpt.

Use in this proof The result is used as the fixed-L acceptance-decay estimate: for each fixed real $\delta < 1$, it supplies the uniform vanishing of $\alpha_{U,\varepsilon,L}(x_0, p_0)$ over the momentum ball $\|p_0\| \le \|x_0\|^\delta$ as $\|x_0\| \to \infty$, together with the equivalent threshold form. In the supplied usage context this estimate is part of the argument controlling the HMC acceptance probability along the state space; no uniformity in L or in δ is asserted.

Lemma 2 (rejection-mass lower bound for the HMC kernel)**.** *Let $d \in \mathbb{N}$, let $\mathcal{X} = \mathbb{R}^d$, and let μ be the standard Gaussian probability measure on $\mathcal{X}$. Let $U : \mathcal{X} \to \mathbb{R}$, $L \in \mathbb{N}$, and $\varepsilon \in \mathbb{R}$. For $(x, p) \in \mathcal{X} \times \mathcal{X}$, write*

$$(q_1(x,p), q_2(x,p)) = \Phi_\varepsilon^L(x,p)$$

for the L-step leapfrog Hamiltonian flow generated by U with step size ε, and define the

acceptance probability

$$\alpha(x,p) = \min\left\{1,\, \exp\left(U(x) - U(q_1(x,p)) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|q_2(x,p)\|^2\right)\right\}.$$

Let P be the HMC kernel

$$P(x,\cdot) = \int_{\mathcal{X}} \big(\alpha(x,p)\delta_{q_1(x,p)} + \big(1-\alpha(x,p)\big)\delta_x\big)\,\mu(dp).$$

Assume the following conditions:

(i) $0 \le \alpha(x,p) \le 1$ *for all* $x, p \in \mathcal{X}$*;*

(ii) for each $x \in \mathcal{X}$*, the map* $p \mapsto \alpha(x,p)$ *is almost everywhere measurable with respect to* μ*;*

(iii) for each $x \in \mathcal{X}$*, the map* $p \mapsto \alpha(x,p)$ *is integrable with respect to* μ*;*

(iv) for each $x \in \mathcal{X}$*, the measure-valued map*

$$p \longmapsto \alpha(x,p)\delta_{q_1(x,p)} + \big(1-\alpha(x,p)\big)\delta_x$$

is almost everywhere measurable with respect to μ*.*

Then for every $x \in \mathcal{X}$ *and every measurable set* $A \subseteq \mathcal{X}$*,*

$$\left(1 - \int_{\mathcal{X}} \alpha(x,p)\,\mu(dp)\right)\delta_x(A) \le P(x,A),$$

where both sides are interpreted as extended nonnegative numbers and $\delta_x(A) = \mathbf{1}_A(x)$*.*

Proof. First, μ has total mass one. Indeed, the standard Gaussian momentum law is the image of a Gaussian probability measure under a measurable linear equivalence, so it is a probability measure. Hence

$$\int_{\mathcal{X}} 1\, d\mu = 1.$$

Fix $x \in \mathcal{X}$, and write $\alpha_x(p) = \alpha(x,p)$. By Condition (ii), α_x is almost everywhere measurable with respect to μ, and by Condition (iii) it is integrable. The constant function 1 is integrable with respect to μ, so $1 - \alpha_x$ is integrable. By Condition (i), $1 - \alpha_x \ge 0$ everywhere, hence almost everywhere. Therefore the ordinary real integral satisfies

$$\int_{\mathcal{X}} \big(1-\alpha_x(p)\big)\,\mu(dp) = \int_{\mathcal{X}} 1\,\mu(dp) - \int_{\mathcal{X}} \alpha_x(p)\,\mu(dp) = 1 - \int_{\mathcal{X}} \alpha_x(p)\,\mu(dp).$$

Let $\iota : [0,\infty) \to [0,\infty]$ denote the natural inclusion. Since $1 - \alpha_x \geq 0$ is integrable, the nonnegative integral of $\iota(1-\alpha_x)$ equals the inclusion of the ordinary integral:

$$\int_{\mathcal{X}} \iota\big(1 - \alpha_x(p)\big)\, \mu(dp) = \iota\left(1 - \int_{\mathcal{X}} \alpha_x(p)\, \mu(dp)\right).$$

Moreover, $p \mapsto \iota(1-\alpha_x(p))$ is almost everywhere measurable, because α_x is almost everywhere measurable and $1 - \alpha_x$ is obtained from it by subtraction from a constant.

By the definition of the HMC kernel,

$$P(x,\cdot) = \int_{\mathcal{X}} \big(\iota(\alpha_x(p))\delta_{q_1(x,p)} + \iota\big(1-\alpha_x(p)\big)\delta_x\big)\, \mu(dp).$$

Thus, using the almost everywhere measurability of the measure-valued integrand from Condition (iv) and the measurability of A, the value of the kernel on A is

$$P(x,A) = \int_{\mathcal{X}} \big(\iota(\alpha_x(p))\delta_{q_1(x,p)}(A) + \iota\big(1-\alpha_x(p)\big)\delta_x(A)\big)\, \mu(dp),$$

where the integral is the nonnegative integral.

For each p, additivity and scalar multiplication for measures give

$$\big(\iota(\alpha_x(p))\delta_{q_1(x,p)} + \iota\big(1-\alpha_x(p)\big)\delta_x\big)\,(A) = \iota(\alpha_x(p))\delta_{q_1(x,p)}(A) + \iota\big(1-\alpha_x(p)\big)\delta_x(A).$$

Since the first term on the right is nonnegative,

$$\iota\big(1-\alpha_x(p)\big)\delta_x(A) \leq \iota(\alpha_x(p))\delta_{q_1(x,p)}(A) + \iota\big(1-\alpha_x(p)\big)\delta_x(A).$$

Integrating this pointwise inequality with respect to μ, and using monotonicity of the nonnegative integral, yields

$$\int_{\mathcal{X}} \iota\big(1-\alpha_x(p)\big)\delta_x(A)\, \mu(dp) \leq \int_{\mathcal{X}} \big(\iota(\alpha_x(p))\delta_{q_1(x,p)}(A) + \iota\big(1-\alpha_x(p)\big)\delta_x(A)\big)\, \mu(dp) = P(x,A).$$

Finally, the factor $\delta_x(A)$, which is either 0 or 1, can be pulled out of the nonnegative integral. Using the almost everywhere measurability of $p \mapsto \iota(1-\alpha_x(p))$,

$$\int_{\mathcal{X}} \iota\big(1-\alpha_x(p)\big)\delta_x(A)\, \mu(dp) = \left(\int_{\mathcal{X}} \iota\big(1-\alpha_x(p)\big)\, \mu(dp)\right)\delta_x(A) = \iota\left(1 - \int_{\mathcal{X}} \alpha_x(p)\, \mu(dp)\right)\delta_x(A).$$

Combining this identity with the previous inequality gives

$$\iota\left(1-\int_{\mathcal{X}}\alpha(x,p)\,\mu(dp)\right)\delta_x(A)\le P(x,A),$$

which is the desired conclusion.

□

Lemma 3 (Measurability and an atom lower bound for the Metropolis-adjusted Hamiltonian kernel)**.** *Let $d\in\mathbb{N}$, let $X_d=\mathbb{R}^d$ be Euclidean space with norm $\|\cdot\|$ and its Borel σ-algebra, and write ∇U for the Euclidean gradient of a differentiable function $U:X_d\to\mathbb{R}$. Let μ_d be the standard Gaussian probability measure on X_d, with density*

$$p\mapsto(2\pi)^{-d/2}\exp\!\left(-\frac{\|p\|^2}{2}\right)$$

with respect to Lebesgue measure. For $L\in\mathbb{N}$ and $\varepsilon\in\mathbb{R}$, define the one-step map $S_{U,\varepsilon}:X_d\times X_d\to X_d\times X_d$ by

$$S_{U,\varepsilon}(x,p)=\Big(x+\varepsilon\big(p-\tfrac{\varepsilon}{2}\nabla U(x)\big),\ p-\tfrac{\varepsilon}{2}\nabla U(x)-\tfrac{\varepsilon}{2}\nabla U\big(x+\varepsilon\big(p-\tfrac{\varepsilon}{2}\nabla U(x)\big)\big)\Big).$$

Define $\Phi^0_{U,\varepsilon}$ to be the identity and $\Phi^{n+1}_{U,\varepsilon}=\Phi^n_{U,\varepsilon}\circ S_{U,\varepsilon}$. For $(x,p)\in X_d\times X_d$, write

$$\Phi^L_{U,\varepsilon}(x,p)=(q_x,q_p).$$

Set

$$\alpha_{U,L,\varepsilon}(x,p)=\min\left\{1,\exp\!\left(U(x)-U(q_x)+\frac{1}{2}\|p\|^2-\frac{1}{2}\|q_p\|^2\right)\right\}.$$

Finally, for each $x\in X_d$ and Borel set $A\subseteq X_d$, define

$$K_{U,L,\varepsilon}(x,A)=\int_{X_d}\big(\alpha_{U,L,\varepsilon}(x,p)\mathbf{1}_A(q_x)+\big(1-\alpha_{U,L,\varepsilon}(x,p)\big)\mathbf{1}_A(x)\big)\,\mu_d(dp).$$

If U is Fréchet differentiable, then:

the measure-valued map

$$x\longmapsto K_{U,L,\varepsilon}(x,\cdot)$$

is measurable with respect to the σ-algebra on Borel measures generated by the evaluation maps $m\mapsto m(A)$ for Borel $A\subseteq X_d$, and for every $x\in X_d$,

$$\left(1-\int_{X_d}\alpha_{U,L,\varepsilon}(x,p)\,\mu_d(dp)\right)\delta_x\le K_{U,L,\varepsilon}(x,\cdot),$$

where the coefficient $1 - \int_{X_d} \alpha_{U,L,\varepsilon}(x,p)\,\mu_d(dp)$ *is nonnegative because* $0 \le \alpha_{U,L,\varepsilon} \le 1$.

Proof. First, differentiability of U implies continuity, hence Borel measurability of U. The Fréchet derivative $x \mapsto DU(x)$ is measurable on the finite-dimensional Euclidean space X_d, and the Riesz identification between X_d^* and X_d is continuous. Therefore the gradient field

$$x \longmapsto \nabla U(x)$$

is measurable.

Now examine the one-step map. On $X_d \times X_d$, write $z = (x,p)$ and set

$$P(z) = p - \frac{\varepsilon}{2}\nabla U(x).$$

This map is measurable because the second coordinate and the composition $\nabla U(x)$ are measurable. Then

$$Q(z) = x + \varepsilon P(z)$$

is measurable. Since $\nabla U(Q(z))$ is the composition of the measurable gradient field with the measurable map Q, the second coordinate

$$R(z) = P(z) - \frac{\varepsilon}{2}\nabla U(Q(z))$$

is also measurable. Hence

$$S_{U,\varepsilon}(z) = (Q(z), R(z))$$

is measurable.

By induction on L, the iterate $\Phi_{U,\varepsilon}^L$ is measurable. For $L = 0$ it is the identity map. For the successor step, the identity

$$\Phi_{U,\varepsilon}^{n+1} = \Phi_{U,\varepsilon}^n \circ S_{U,\varepsilon}$$

expresses $\Phi_{U,\varepsilon}^{n+1}$ as a composition of two measurable maps, so measurability is preserved.

Next, the coordinate projections of $\Phi_{U,\varepsilon}^L(x,p)$, namely q_x and q_p, are measurable functions of (x,p). Therefore the norms $\|p\|$, $\|q_p\|$, and their squares are measurable, as are the terms

$$U(x), \qquad U(q_x), \qquad \frac{1}{2}\|p\|^2, \qquad \frac{1}{2}\|q_p\|^2.$$

It follows that the exponent

$$U(x) - U(q_x) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|q_p\|^2$$

is measurable, so its exponential is measurable. Taking the minimum with the constant 1 preserves measurability. Thus

$$(x,p) \longmapsto \alpha_{U,L,\varepsilon}(x,p)$$

is jointly measurable.

The standard Gaussian momentum measure μ_d is a probability measure. This will be used when integrating the constant function 1.

For a Borel set $A \subseteq X_d$, consider the measure-valued map

$$z = (x,p) \longmapsto \alpha_{U,L,\varepsilon}(x,p)\delta_{q_x} + \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\delta_x.$$

We claim that this map is measurable. It suffices to show that its evaluation at every Borel set A is measurable. For such A, the indicator $\mathbf{1}_A$ is Borel measurable, and

$$\big(\alpha_{U,L,\varepsilon}(x,p)\delta_{q_x} + \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\delta_x\big)\,(A) = \alpha_{U,L,\varepsilon}(x,p)\mathbf{1}_A(q_x) + \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\mathbf{1}_A(x).$$

The first term is measurable because $\alpha_{U,L,\varepsilon}$ and q_x are measurable; the second is measurable because $\alpha_{U,L,\varepsilon}$ and the projection $(x,p) \mapsto x$ are measurable. Hence the evaluation map is measurable for every Borel A, and therefore the measure-valued map is measurable.

For each fixed x, the map $p \mapsto (x,p)$ is measurable, and therefore

$$p \longmapsto \alpha_{U,L,\varepsilon}(x,p)\delta_{q_x} + \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\delta_x$$

is almost everywhere measurable with respect to μ_d. This is precisely the measurability condition needed when forming the measure obtained by integrating this family against μ_d.

We now prove measurability of $x \mapsto K_{U,L,\varepsilon}(x,\cdot)$. Fix a Borel set $A \subseteq X_d$. By the definition of $K_{U,L,\varepsilon}$,

$$K_{U,L,\varepsilon}(x,A) = \int_{X_d} \big(\alpha_{U,L,\varepsilon}(x,p)\mathbf{1}_A(q_x) + \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\mathbf{1}_A(x)\big)\,\mu_d(dp).$$

The integrand is jointly measurable in (x,p). Therefore the nonnegative integral as a function of x is measurable. Thus $x \mapsto K_{U,L,\varepsilon}(x,A)$ is measurable for every Borel A. Since the σ-algebra on measures is generated by these evaluation maps, the measure-valued map $x \mapsto K_{U,L,\varepsilon}(x,\cdot)$ is measurable.

It remains to establish the lower bound at the atom $\{x\}$. Fix $x \in X_d$. Since $\alpha_{U,L,\varepsilon}$ is jointly measurable, the function

$$p \longmapsto \alpha_{U,L,\varepsilon}(x,p)$$

is measurable. Moreover, for every p,

$$0 \leq \alpha_{U,L,\varepsilon}(x,p) \leq 1,$$

because the exponential is positive and the minimum with 1 is taken. Hence

$$0 \leq 1 - \alpha_{U,L,\varepsilon}(x,p) \leq 1.$$

The function $p \mapsto \alpha_{U,L,\varepsilon}(x,p)$ is integrable with respect to μ_d, being measurable and bounded by 1 on a probability space. The constant function 1 is also integrable, with

$$\int_{X_d} 1 \, \mu_d(dp) = 1.$$

Therefore

$$\int_{X_d} \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big) \, \mu_d(dp) = 1 - \int_{X_d} \alpha_{U,L,\varepsilon}(x,p) \, \mu_d(dp).$$

Since $1 - \alpha_{U,L,\varepsilon}(x,p) \geq 0$ for every p, the ordinary real integral agrees with the corresponding nonnegative integral. Thus

$$1 - \int_{X_d} \alpha_{U,L,\varepsilon}(x,p) \, \mu_d(dp) = \int_{X_d} \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big) \, \mu_d(dp).$$

Now apply the definition of $K_{U,L,\varepsilon}$ to the singleton $\{x\}$. Since $\{x\}$ is Borel and the family of measures is almost everywhere measurable,

$$K_{U,L,\varepsilon}(x,\{x\}) = \int_{X_d} \big(\alpha_{U,L,\varepsilon}(x,p)\delta_{q_x}(\{x\}) + \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\delta_x(\{x\})\big) \, \mu_d(dp).$$

For each p, the point x belongs to $\{x\}$, so

$$\delta_x(\{x\}) = 1.$$

Therefore

$$\big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\delta_x(\{x\}) = 1 - \alpha_{U,L,\varepsilon}(x,p).$$

The term

$$\alpha_{U,L,\varepsilon}(x,p)\delta_{q_x}(\{x\})$$

is nonnegative. Hence, for every p,

$$\alpha_{U,L,\varepsilon}(x,p)\delta_{q_x}(\{x\}) + \big(1 - \alpha_{U,L,\varepsilon}(x,p)\big)\delta_x(\{x\}) \geq 1 - \alpha_{U,L,\varepsilon}(x,p).$$

Integrating this pointwise inequality gives

$$K_{U,L,\varepsilon}(x,\{x\}) \geq \int_{X_d} \big(1-\alpha_{U,L,\varepsilon}(x,p)\big)\,\mu_d(dp) = 1-\int_{X_d} \alpha_{U,L,\varepsilon}(x,p)\,\mu_d(dp).$$

This proves the desired lower bound on the atom $\{x\}$.

Finally, let $A \subseteq X_d$ be any Borel set and put

$$c_x = 1-\int_{X_d} \alpha_{U,L,\varepsilon}(x,p)\,\mu_d(dp).$$

The coefficient c_x is nonnegative, because $0 \leq \alpha_{U,L,\varepsilon} \leq 1$. The measure $c_x\delta_x$ evaluates as

$$(c_x\delta_x)(A) = \begin{cases} c_x, & x \in A, \\ 0, & x \notin A. \end{cases}$$

If $x \in A$, then $\{x\} \subseteq A$, and monotonicity of $K_{U,L,\varepsilon}(x,\cdot)$ together with the atom estimate yields

$$(c_x\delta_x)(A) = c_x \leq K_{U,L,\varepsilon}(x,\{x\}) \leq K_{U,L,\varepsilon}(x,A).$$

If $x \notin A$, then

$$(c_x\delta_x)(A) = 0 \leq K_{U,L,\varepsilon}(x,A).$$

Thus

$$c_x\delta_x \leq K_{U,L,\varepsilon}(x,\cdot)$$

as measures. Since x was arbitrary, the required lower bound holds for every $x \in X_d$. This completes the proof.

□

Lemma 4. *Let d be a positive integer, let $\mathbb{R}^d$ be equipped with the Euclidean norm $\|\cdot\|$, and let λ_d denote Lebesgue measure on $\mathbb{R}^d$. Let $U : \mathbb{R}^d \to \mathbb{R}$ be Borel measurable, let π be a measure on $\mathbb{R}^d$, let $R \in \mathbb{R}$, and let $c \in [0,\infty]$ satisfy*

(i) $c \neq 0$ and $c \neq \infty$, so $0 < c < \infty$;

(ii) for every Borel set $A \subseteq \mathbb{R}^d$,

$$\pi(A) = c\int_A e^{-U(x)}\,d\lambda_d(x),$$

equivalently π is the measure obtained by multiplying by c the Lebesgue measure with density $x \mapsto e^{-U(x)}$.

Then

$$\pi\big(\{x \in \mathbb{R}^d : R < \|x\|\}\big) > 0.$$

Proof. Set

$$S_R = \{x \in \mathbb{R}^d : R < \|x\|\}, \qquad f(x) = e^{-U(x)}.$$

The function f is Borel measurable because U is measurable, exponentiation is continuous, and f is the composition of measurable functions.

The set S_R is open because the norm is continuous, and therefore it is Borel measurable.

Since $d \geq 1$, choose a nonzero vector $y \in \mathbb{R}^d$; for instance, one may take a coordinate unit vector. Then $\|y\| > 0$. To see that S_R is nonempty, define

$$z = \frac{\max(R, 0) + 1}{\|y\|}\, y.$$

The scalar $(\max(R, 0) + 1)/\|y\|$ is positive, and hence

$$\|z\| = \frac{\max(R, 0) + 1}{\|y\|}\, \|y\| = \max(R, 0) + 1.$$

Since $\max(R, 0) \geq R$, we have

$$R < \max(R, 0) + 1 = \|z\|.$$

Thus $z \in S_R$, so S_R is nonempty.

The set S_R is therefore a nonempty open subset of $\mathbb{R}^d$. Since Lebesgue measure is positive on nonempty open sets, it follows that

$$\lambda_d(S_R) > 0.$$

Next, $f(x) = e^{-U(x)} > 0$ for every x, so the support of f is all of $\mathbb{R}^d$. Applying the criterion for positivity of the integral of a nonnegative measurable function in terms of the measure of its support, with the measure taken to be Lebesgue measure restricted to S_R, gives

$$I_R := \int_{S_R} e^{-U(x)}\, d\lambda_d(x) > 0.$$

Indeed, the support intersects S_R in all of S_R, and $\lambda_d(S_R) > 0$.

Let

$$\nu(A) = \int_A e^{-U(x)}\, d\lambda_d(x)$$

be the Lebesgue measure with density f. Since S_R is measurable, the definition of the density

measure gives

$$\nu(S_R) = \int_{S_R} e^{-U(x)}\, d\lambda_d(x) = I_R.$$

By assumption, $\pi = c\nu$, and therefore

$$\pi(S_R) = c\,\nu(S_R) = c\,I_R.$$

Now $c > 0$ and $I_R > 0$. Hence the extended nonnegative product cI_R is positive, so

$$\pi\big(\{x \in \mathbb{R}^d : R < \|x\|\}\big) > 0.$$

This proves the claim.

□

Lemma 5. *Let $d \in \mathbb{N}$, and let $E_d = \mathbb{R}^d$ be Euclidean space with its usual inner product. For every differentiable function $U : E_d \to \mathbb{R}$, define its gradient $\nabla U : E_d \to E_d$ by the characterizing relation*

$$\langle \nabla U(x), v\rangle = DU(x)(v) \qquad \text{for all } v \in E_d,$$

where $DU(x)$ is the Fréchet derivative of U at x. Then ∇U is Borel measurable.

Proof. Let E_d^* denote the continuous dual space of E_d. Consider the Riesz representation map

$$\Phi : E_d \to E_d^*, \qquad \Phi(w)(v) = \langle w, v\rangle.$$

This map is a linear homeomorphism. Because U is differentiable, the Fréchet derivative map

$$x \mapsto DU(x)$$

is a Borel measurable map from E_d into E_d^*. Since Φ^{-1} is continuous, it is Borel measurable. For every x, the defining property of the gradient gives

$$\Phi(\nabla U(x)) = DU(x),$$

hence

$$\nabla U(x) = \Phi^{-1}\big(DU(x)\big).$$

Thus ∇U is the composition of the measurable map $x \mapsto DU(x)$ with the measurable map Φ^{-1}. Therefore ∇U is Borel measurable, as claimed.

□

Lemma 6 (Measurability and probability of the HMC transition kernel)**.** *Let $d \in \mathbb{N}$, let $U : \mathbb{R}^d \to \mathbb{R}$ be differentiable, let $L \in \mathbb{N}$, and let $\varepsilon \in \mathbb{R}$. For $(x, p) \in \mathbb{R}^d \times \mathbb{R}^d$, define one leapfrog step by*

$$p_{1/2} = p - \frac{\varepsilon}{2}\nabla U(x), \qquad x' = x + \varepsilon p_{1/2}, \qquad p' = p_{1/2} - \frac{\varepsilon}{2}\nabla U(x'),$$

and set $S_\varepsilon(x, p) = (x', p')$. Let $\Phi_{L,\varepsilon} = S_\varepsilon^{\circ L}$ be the L-fold iterate, and write

$$(x_L, p_L) = \Phi_{L,\varepsilon}(x, p).$$

Define the acceptance probability

$$\alpha(x, p) = \min\left\{1, \exp\left(U(x) - U(x_L) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|p_L\|^2\right)\right\}.$$

Let $\mu_G = N(0, I_d)$ be the standard Gaussian momentum law on $\mathbb{R}^d$. For each $x \in \mathbb{R}^d$, define a measure $P(x, \cdot)$ by

$$P(x, \cdot) = \int_{\mathbb{R}^d} \left(\alpha(x, p)\, \delta_{x_L(x,p)} + (1 - \alpha(x, p))\, \delta_x\right) \mu_G(dp),$$

where δ_y denotes the Dirac mass at y. Then:

(i) for every $x \in \mathbb{R}^d$, the function $p \mapsto \alpha(x, p)$ is Borel measurable;

(ii) for every $x \in \mathbb{R}^d$, $P(x, \cdot)$ is a probability measure on $\mathbb{R}^d$.

Proof. Fix d, U, L, ε, and assume that U is differentiable. By Lemma 5, the gradient ∇U is Borel measurable.

We first show that the leapfrog step S_ε is measurable on $\mathbb{R}^d \times \mathbb{R}^d$. The coordinate projections $(x, p) \mapsto x$ and $(x, p) \mapsto p$ are measurable. Since ∇U is measurable, the map $(x, p) \mapsto \nabla U(x)$ is measurable. Hence

$$(x, p) \mapsto p - \frac{\varepsilon}{2}\nabla U(x)$$

is measurable. Consequently,

$$(x, p) \mapsto x + \varepsilon\left(p - \frac{\varepsilon}{2}\nabla U(x)\right)$$

is measurable; this is the first coordinate x' of $S_\varepsilon(x, p)$. Using this expression for x', the map

$$(x, p) \mapsto \left(p - \frac{\varepsilon}{2}\nabla U(x)\right) - \frac{\varepsilon}{2}\nabla U(x')$$

is also measurable; this is the second coordinate p'. Therefore the pair (x', p') is a measurable map, so S_ε is measurable. By induction on L, the L-fold iterate $\Phi_{L,\varepsilon}$ is measurable.

For assertion (i), fix $x \in \mathbb{R}^d$. The map $p \mapsto (x, p)$ is measurable, and since $\Phi_{L,\varepsilon}$ is measurable, the map

$$p \mapsto q(p) := \Phi_{L,\varepsilon}(x, p)$$

is measurable. Its two coordinates $q_1(p) = x_L(x, p)$ and $q_2(p) = p_L(x, p)$ are measurable. Because U is differentiable, it is continuous, hence Borel measurable; therefore $p \mapsto U(q_1(p))$ is measurable. The maps $p \mapsto \|p\|^2$ and $p \mapsto \|q_2(p)\|^2$ are also measurable, since the norm is continuous and q_2 is measurable. Thus

$$p \mapsto U(x) - U(q_1(p)) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|q_2(p)\|^2$$

is measurable. The real exponential is continuous, and taking the minimum with the constant 1 preserves measurability. Hence $p \mapsto \alpha(x, p)$ is measurable. This proves (i).

For assertion (ii), fix $x \in \mathbb{R}^d$. For each $p \in \mathbb{R}^d$, set

$$K_x(p) = \alpha(x, p)\, \delta_{x_L(x,p)} + (1 - \alpha(x, p))\, \delta_x.$$

We first show that the measure-valued map $p \mapsto K_x(p)$ is measurable. Let $E \subseteq \mathbb{R}^d$ be an arbitrary Borel set. For a Dirac mass, $\delta_y(E) = \mathbf{1}_E(y)$. Therefore

$$K_x(p)(E) = \alpha(x, p)\, \mathbf{1}_E(x_L(x, p)) + (1 - \alpha(x, p))\, \mathbf{1}_E(x).$$

The function $p \mapsto \alpha(x, p)$ is measurable by part (i), and $p \mapsto \mathbf{1}_E(x_L(x, p))$ is measurable because $x_L(x, \cdot)$ is measurable. Hence the first summand is measurable. The second summand is measurable because $\alpha(x, \cdot)$ is measurable and x is fixed. Thus $p \mapsto K_x(p)(E)$ is measurable for every Borel set E, so $p \mapsto K_x(p)$ is a measurable family of measures.

Next we check that each $K_x(p)$ has total mass 1. Put

$$a = \alpha(x, p) = \min\left\{1, \exp\left(U(x) - U(x_L(x, p)) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|p_L(x, p)\|^2\right)\right\}.$$

Since the exponential is strictly positive, $a > 0$, and by definition $a \leq 1$. Hence $0 \leq a \leq 1$, and in particular $1 - a \geq 0$. The Dirac masses each have total mass 1, so

$$K_x(p)(\mathbb{R}^d) = a\, \delta_{x_L(x,p)}(\mathbb{R}^d) + (1 - a)\, \delta_x(\mathbb{R}^d) = a + (1 - a) = 1.$$

Thus every measure $K_x(p)$ is a probability measure.

The standard Gaussian momentum law $\mu_G = N(0, I_d)$ is a probability measure: it is the image of the product of d standard Gaussian probability measures under the coordinate identification with $\mathbb{R}^d$. Now the total mass of $P(x, \cdot)$ is obtained by integrating the total masses of the measurable family $K_x(p)$ with respect to μ_G:

$$P(x, \mathbb{R}^d) = \int_{\mathbb{R}^d} K_x(p)(\mathbb{R}^d)\, \mu_G(dp) = \int_{\mathbb{R}^d} 1\, \mu_G(dp) = 1.$$

Therefore $P(x, \cdot)$ has total mass 1, so it is a probability measure. This proves (ii).

□

Lemma 7 (Iterated HMC kernels dominate the rejection-mass atom)**.** *Let $d \in \mathbb{N}$, $U \colon \mathbb{R}^d \to \mathbb{R}$, $L \in \mathbb{N}$, and $\varepsilon \in \mathbb{R}$. Let μ_d be the standard Gaussian probability measure on the momentum space $\mathbb{R}^d$. For $(x, p) \in \mathbb{R}^d \times \mathbb{R}^d$, let*

$$(q_1, q_2) = \Phi_{L,\varepsilon}(x, p)$$

be the endpoint obtained after L leapfrog steps of size ε applied to the Hamiltonian $U(x) + \frac{1}{2}\|p\|^2$, and define the acceptance probability

$$a(x, p) = \min\left\{1,\, \exp\left(U(x) - U(q_1) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|q_2\|^2\right)\right\}.$$

Let $K(x, \cdot)$ denote the HMC transition kernel

$$K(x, \cdot) = \int_{\mathbb{R}^d} \left(a(x, p)\, \delta_{q_1(x,p)} + (1 - a(x, p))\, \delta_x\right)\, \mu_d(dp),$$

and let $K^n(x, \cdot)$ denote its n-fold iterate, with $K^0(x, \cdot) = \delta_x$. Put

$$r(x) = \int_{\mathbb{R}^d} a(x, p)\, \mu_d(dp), \qquad R(x) = 1 - r(x).$$

Assume:

(i) for every $x \in \mathbb{R}^d$,

$$R(x)\, \delta_x \le K(x, \cdot);$$

(ii) for every $x \in \mathbb{R}^d$, $0 \le R(x)$;

(iii) for every $x \in \mathbb{R}^d$, $R(x) \le 1$;

(iv) the map $x \mapsto K(x, \cdot)$ is measurable.

Then for every $n \in \mathbb{N}$ and every $x \in \mathbb{R}^d$,

$$R(x)^n \, \delta_x \leq K^n(x, \cdot).$$

Proof. We prove the assertion by induction on n.

For the base case $n = 0$, we have $K^0(x, \cdot) = \delta_x$ and $R(x)^0 = 1$, so

$$R(x)^0 \delta_x = \delta_x = K^0(x, \cdot).$$

Thus the claim holds for $n = 0$.

Now assume the claim for some n and all x, and fix $x \in \mathbb{R}^d$. Write

$$R = R(x) = 1 - \int_{\mathbb{R}^d} a(x, p) \, \mu_d(dp).$$

By condition (ii), $R \geq 0$. By condition (i),

$$R \, \delta_x \leq K(x, \cdot).$$

The induction hypothesis gives

$$R^n \, \delta_x \leq K^n(x, \cdot).$$

We compare through the intermediate measure

$$(R^n \, \delta_x) \operatorname{bind} K.$$

First, for every measurable set $A \subseteq \mathbb{R}^d$,

$$((R^n \, \delta_x) \operatorname{bind} K)(A) = \int_{\mathbb{R}^d} K(y, A) \, d(R^n \, \delta_x)(y) = R^n \int_{\mathbb{R}^d} K(y, A) \, d\delta_x(y) = R^n K(x, A).$$

Here the first equality is the defining formula for the composition of a measure with a transition kernel; condition (iv) supplies the required measurability in the variable y. On the other hand,

$$(R^{n+1} \delta_x)(A) = R^n (R \delta_x)(A).$$

Since $R\delta_x \leq K(x, \cdot)$, we have

$$(R\delta_x)(A) \leq K(x, A).$$

Multiplying by $R^n \geq 0$, we obtain

$$(R^{n+1}\delta_x)(A) \leq R^n K(x, A) = ((R^n\,\delta_x)\,\text{bind}\,K)\,(A).$$

Because this holds for every measurable A,

$$R^{n+1}\delta_x \leq (R^n\,\delta_x)\,\text{bind}\,K.$$

Next, we use monotonicity of composition with the kernel K. If $m_1 \leq m_2$ are measures on $\mathbb{R}^d$, then for every measurable $A \subseteq \mathbb{R}^d$,

$$(m_1\,\text{bind}\,K)(A) = \int_{\mathbb{R}^d} K(y, A)\,dm_1(y) \leq \int_{\mathbb{R}^d} K(y, A)\,dm_2(y) = (m_2\,\text{bind}\,K)(A),$$

by monotonicity of the integral with respect to the underlying measure. Applying this to the induction hypothesis

$$R^n\,\delta_x \leq K^n(x, \cdot)$$

yields

$$(R^n\,\delta_x)\,\text{bind}\,K \leq K^n(x, \cdot)\,\text{bind}\,K = K^{n+1}(x, \cdot),$$

where the last equality is the recursive definition of the iterated kernel.

Combining the two inequalities gives

$$R^{n+1}\delta_x \leq K^{n+1}(x, \cdot).$$

Thus the claim holds for $n + 1$. By induction, it holds for every $n \in \mathbb{N}$ and every $x \in \mathbb{R}^d$.

□

Lemma 8 (Large-position rejection probability)**.** *For each $d \in \mathbb{N}$, write $E_d = \mathbb{R}^d$ with its Euclidean norm $\|\cdot\|$, and let μ_d be the standard Gaussian momentum law on E_d, namely the pushforward of the product law $\bigotimes_{i=1}^d N(0, 1)$ under the coordinate identification. Fix a potential $U : E_d \to \mathbb{R}$, an integer $L \in \mathbb{N}$, and a step size $\varepsilon \in \mathbb{R}$. For $(x, p) \in E_d \times E_d$, let $(q_1, q_2) = \Phi_{L,\varepsilon}(x, p)$ be the point obtained by applying L leapfrog steps of size ε to the Hamiltonian $U(z) + \frac{1}{2}\|w\|^2$, and define the acceptance probability*

$$\alpha(x, p) = \min\left\{1,\, \exp\left(U(x) - U(q_1) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|q_2\|^2\right)\right\}.$$

Assume:

(i) $0 \leq \alpha(x, p)$ *for all* $x, p \in E_d$*;*

(ii) $\alpha(x,p) \le 1$ *for all* $x, p \in E_d$*;*

(iii) for every $x \in E_d$*, the function* $p \mapsto \alpha(x,p)$ *is measurable with respect to* μ_d*;*

(iv) for every $\eta > 0$ *there exists* $R_0 \in \mathbb{R}$ *such that for all* $x \in E_d$ *with* $R_0 < \|x\|$ *and all* $p \in E_d$ *with* $\|p\| \le \sqrt{\|x\|}$*, one has* $\alpha(x,p) \le \eta$*.*

Let $\gamma \in \mathbb{R}$ *satisfy*

(v) $0 < \gamma < 1$*.*

Then there exists $R \in \mathbb{R}$ *such that for every* $x \in E_d$ *with* $R < \|x\|$*,*

$$1 - \gamma \le 1 - \int_{E_d} \alpha(x,p)\, d\mu_d(p).$$

Proof. Set $\mu = \mu_d$. By construction, μ is the pushforward of a product of probability measures under a measurable coordinate identification, so μ is a probability measure on E_d. In particular, $\mu(E_d) = 1$. Since $0 < \gamma < 1$, we have $\gamma/2 > 0$.

We first prove a tail estimate for the momentum norm. For $n \in \mathbb{N}$, define

$$S_n = \{p \in E_d : n < \|p\|\}.$$

Each S_n is measurable. The family is decreasing in n: if $m \le n$, then $S_n \subseteq S_m$. Moreover,

$$\bigcap_{n \in \mathbb{N}} S_n = \varnothing,$$

because for every p there is an integer $n \ge \|p\|$, and then $p \notin S_n$. Also $\mu(S_0) \le \mu(E_d) = 1 < \infty$. Therefore, by continuity from above for a decreasing sequence of measurable sets of finite measure,

$$\lim_{n \to \infty} \mu(S_n) = \mu\left(\bigcap_{n \in \mathbb{N}} S_n\right) = \mu(\varnothing) = 0.$$

Since $\gamma/2 > 0$, it follows that for all sufficiently large n,

$$\mu(S_n) \le \frac{\gamma}{2}.$$

Choose one such $N \in \mathbb{N}$, so that

$$\mu\{p \in E_d : N < \|p\|\} \le \frac{\gamma}{2}.$$

Now apply condition (iv) with $\eta = \gamma/2$. We obtain $R_0 \in \mathbb{R}$ such that for every $x \in E_d$ with $R_0 < \|x\|$ and every $p \in E_d$ with $\|p\| \le \sqrt{\|x\|}$,

$$\alpha(x,p) \le \frac{\gamma}{2}.$$

Define

$$R = \max\{R_0, N^2\}.$$

We claim that this R satisfies the required conclusion.

Fix $x \in E_d$ with $R < \|x\|$. Then $R_0 < \|x\|$ and $N^2 < \|x\|$. Since $N \ge 0$, taking square roots gives

$$N < \sqrt{\|x\|}.$$

Let

$$A = \{p \in E_d : \|p\| \le \sqrt{\|x\|}\}.$$

This set is measurable. By conditions (i)(iii), the function $p \mapsto \alpha(x,p)$ is measurable, nonnegative, and bounded above by 1. Since μ is a probability measure, this function is integrable with respect to μ. Hence the integral can be split as

$$\int_{E_d} \alpha(x,p)\, d\mu(p) = \int_A \alpha(x,p)\, d\mu(p) + \int_{A^c} \alpha(x,p)\, d\mu(p).$$

For $p \in A$, condition (iv) gives $\alpha(x,p) \le \gamma/2$. Therefore, by monotonicity of the integral,

$$\int_A \alpha(x,p)\, d\mu(p) \le \int_A \frac{\gamma}{2}\, d\mu(p) = \frac{\gamma}{2}\,\mu(A) \le \frac{\gamma}{2},$$

because $\mu(A) \le \mu(E_d) = 1$.

For the complement, if $p \in A^c$, then $\|p\| > \sqrt{\|x\|} > N$, so $p \in S_N$. Thus $A^c \subseteq S_N$, and consequently

$$\mu(A^c) \le \mu(S_N) \le \frac{\gamma}{2}.$$

On A^c, condition (ii) gives $\alpha(x,p) \le 1$. Hence

$$\int_{A^c} \alpha(x,p)\, d\mu(p) \le \int_{A^c} 1\, d\mu(p) = \mu(A^c) \le \frac{\gamma}{2}.$$

Combining the two estimates,

$$\int_{E_d} \alpha(x,p)\, d\mu(p) \le \frac{\gamma}{2} + \frac{\gamma}{2} = \gamma.$$

Therefore

$$1 - \gamma \leq 1 - \int_{E_d} \alpha(x, p)\, d\mu(p).$$

Since x was arbitrary with $\|x\| > R$, the chosen R verifies the required conclusion.

□

Lemma 9 (Null singletons for measures with an exponential density)**.** *Let $d \in \mathbb{N}$, let $\mathcal{X}_d = \mathbb{R}^d$ be equipped with Lebesgue measure* vol*, let $U : \mathcal{X}_d \to \mathbb{R}$, and let π be a measure on $\mathcal{X}_d$. Suppose that:*

(i) there exists $c \in (0, \infty)$ such that

$$\pi(A) = c \int_A e^{-U(y)}\, \mathrm{vol}(dy)$$

for every measurable $A \subseteq \mathcal{X}_d$;

(ii) the density $y \mapsto e^{-U(y)}$ is measurable;

(iii) for every $y \in \mathcal{X}_d$, the Lebesgue measure of the singleton $\{y\}$ is zero.

Then for every $x \in \mathcal{X}_d$,

$$\pi(\{x\}) = 0.$$

Proof. Fix $x \in \mathcal{X}_d$. By Condition (i), choose $c \in (0, \infty)$ and write

$$\nu(A) = \int_A e^{-U(y)}\, \mathrm{vol}(dy)$$

for measurable $A \subseteq \mathcal{X}_d$, so that $\pi = c\,\nu$.

We first show that $\nu(\{x\}) = 0$. Since the density $g(y) = e^{-U(y)}$ is measurable by Condition (ii), the value of the density measure on a set is governed by the underlying Lebesgue measure of the part of that set on which the density is nonzero. For the singleton $\{x\}$, that part is contained in $\{x\}$. By Condition (iii), $\mathrm{vol}(\{x\}) = 0$, hence the Lebesgue measure of that part is also zero. Therefore

$$\nu(\{x\}) = 0.$$

Equivalently, $\nu(\{x\}) = \int_{\{x\}} g(y)\, \mathrm{vol}(dy) = 0$, since the domain of integration has Lebesgue measure zero.

Finally, using the scalar multiple representation of π,

$$\pi(\{x\}) = (c\,\nu)(\{x\}) = c\,\nu(\{x\}) = c \cdot 0 = 0.$$

Since $x \in \mathcal{X}_d$ was arbitrary, $\pi(\{x\}) = 0$ for every $x \in \mathcal{X}_d$.

□

Lemma 10 (Total-variation lower bound from iterated rejection mass)**.** *Let $d \in \mathbb{N}$, let X_d be the Hamiltonian Monte Carlo state space equipped with its Borel measurable structure, and let $P(x, \cdot)$ be the Hamiltonian Monte Carlo transition kernel with potential $U : X_d \to \mathbb{R}$, L leapfrog steps, and step size ε. Define the iterated kernels by*

$$P^0(x, \cdot) = \delta_x, \qquad P^{k+1}(x, \cdot) = (P^k(x, \cdot))P,$$

where $(\nu P)(A) = \int P(y, A)\, \nu(dy)$. For finite measures μ, ν on X_d, define

$$d_{\mathrm{TV}}(\mu, \nu) = \sup \left\{ \left| \int f \, d\mu - \int f \, d\nu \right| : f : X_d \to \mathbb{R} \text{ measurable, } |f| \leq 1 \right\}.$$

Let π be a finite measure on X_d, and let $r : X_d \to \mathbb{R}$. Assume:

(i) $\pi(\{x\}) = 0$ *for every* $x \in X_d$*;*

(ii) for every $k \in \mathbb{N}$ *and every* $y \in X_d$*,*

$$(r(y))^k \, \delta_y \leq P^k(y, \cdot)$$

as measures on X_d;

(iii) $r(y) \geq 0$ *for every* $y \in X_d$*;*

(iv) $r(y) \leq 1$ *for every* $y \in X_d$*;*

(v) $P(y, \cdot)$ is a probability measure for every $y \in X_d$;

(vi) π is finite;

(vii) every singleton $\{y\}$, $y \in X_d$, is measurable.

Then for every $n \in \mathbb{N}$ and every $x \in X_d$,

$$(r(x))^n \leq d_{\mathrm{TV}}\big(P^n(x, \cdot), \pi\big).$$

Proof. Fix $n \in \mathbb{N}$ and $x \in X_d$, and set

$$m := P^n(x, \cdot).$$

We first show that every iterated kernel has total mass at most 1. For $k = 0$,

$$P^0(x, X_d) = \delta_x(X_d) = 1.$$

If $P^k(x, X_d) \leq 1$, then, using that $P(y, X_d) = 1$ for all y by condition (v),

$$P^{k+1}(x, X_d) = \int_{X_d} P(y, X_d)\, P^k(x, dy) \leq \int_{X_d} 1\, P^k(x, dy) = P^k(x, X_d) \leq 1.$$

Thus $m(X_d) \leq 1$, so m is a finite measure. In particular $m(\{x\}) < \infty$.

By condition (vii), the singleton $\{x\}$ is measurable. Evaluating the measure inequality of condition (ii) at $k = n$ on $\{x\}$ gives

$$(r(x))^n = \big((r(x))^n \delta_x\big)(\{x\}) \leq m(\{x\}).$$

Together with condition (iii), this also shows $m(\{x\}) \geq 0$.

Now define

$$g(y) := \mathbf{1}_{\{x\}}(y).$$

This function is measurable by condition (vii), and $|g(y)| \leq 1$ for every y. Since m and π are finite, the corresponding real integrals are well defined. Using the indicator formula,

$$\int_{X_d} g\, dm = m(\{x\}),$$

and, by condition (i),

$$\int_{X_d} g\, d\pi = \pi(\{x\}) = 0.$$

Therefore

$$\left| \int_{X_d} g\, dm - \int_{X_d} g\, d\pi \right| = m(\{x\}).$$

Consequently,

$$(r(x))^n \leq \left| \int_{X_d} g\, dm - \int_{X_d} g\, d\pi \right|.$$

It remains to compare this test-function value with the total variation supremum. Consider the set

$$S = \left\{ \left| \int f\, dm - \int f\, d\pi \right| : f : X_d \to \mathbb{R} \text{ measurable},\ |f| \leq 1 \right\}.$$

For any admissible f in this set, the bound $|f| \le 1$ gives

$$\left|\int_{X_d} f\,dm\right| \le m(X_d), \qquad \left|\int_{X_d} f\,d\pi\right| \le \pi(X_d).$$

Hence, by the triangle inequality,

$$\left|\int_{X_d} f\,dm - \int_{X_d} f\,d\pi\right| \le m(X_d) + \pi(X_d).$$

Thus S is bounded above. Since g is measurable and satisfies $|g| \le 1$, the value

$$\left|\int_{X_d} g\,dm - \int_{X_d} g\,d\pi\right|$$

belongs to S. Therefore it is at most $\sup S$, which is exactly $d_{\mathrm{TV}}(m,\pi)$. Combining the inequalities,

$$(r(x))^n \le \left|\int_{X_d} g\,dm - \int_{X_d} g\,d\pi\right| \le \sup S = d_{\mathrm{TV}}\big(P^n(x,\cdot),\pi\big).$$

Since n and x were arbitrary, the conclusion follows.

□

Main Theorem

Theorem 11 (Non-geometric ergodicity for the HMC kernel)**.** *Let $d \in \mathbb{N}$, let $X = \mathbb{R}^d$ be equipped with the Euclidean norm $\|\cdot\|$, and let $U : X \to \mathbb{R}$, let π be a finite Borel measure on X, let $L \in \mathbb{N}$, and let $\varepsilon \in \mathbb{R}$.*

Let μ_G denote the standard Gaussian momentum law on X. For $x, p \in X$, let

$$(q_1, q_2) = \Phi_{U,L,\varepsilon}(x,p)$$

be the endpoint of L leapfrog steps of size ε, and define the Metropolis acceptance probability

$$\alpha(x,p) = \min\left\{1,\, \exp\left(U(x) - U(q_1) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|q_2\|^2\right)\right\}.$$

Let P_x be the HMC transition probability

$$P_x(A) = \int_X \big[\alpha(x,p)\mathbf{1}_A(q_1(x,p)) + \big(1-\alpha(x,p)\big)\mathbf{1}_A(x)\big]\ \mu_G(dp), \qquad A \in \mathcal{B}(X).$$

For $n \geq 0$, write P_x^n for the n-step transition kernel starting from x, and define the total variation distance between finite Borel measures by

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{\substack{f \text{ Borel} \\ |f| \leq 1}} \left| \int f \, d\mu - \int f \, d\nu \right|.$$

We say that P is geometrically ergodic with respect to π if there exist a real number ρ with $0 \leq \rho < 1$ and a Borel function $M : X \to [0, \infty]$, finite π-almost everywhere, such that, in the extended nonnegative sense,

$$\|P_x^n - \pi\|_{\mathrm{TV}} \leq M(x)\rho^n \qquad \text{for all } n \geq 0 \text{ and all } x \in X.$$

Assume the following conditions:

(i) $d > 0$;

(ii) $U(x) \geq 0$ *for every* $x \in X$;

(iii) U *is differentiable on* X;

(iv) π *is a finite measure;*

(v) there exist a constant c *with* $0 < c < \infty$ *such that*

$$\pi(A) = c \int_A e^{-U(x)} \, dx \qquad \text{for every Borel set } A \subseteq X;$$

(vi) for every $A \in \mathbb{R}$ *there exists* $R \in \mathbb{R}$ *such that, for every* $x \in X$,

$$R < \|x\| \quad \Longrightarrow \quad A < \frac{\|\nabla U(x)\|}{\|x\|};$$

(vii) there exists $C \in \mathbb{R}$ *such that, for all* $x, y \in X$,

$$3\|x\| \leq \|y\| \quad \text{and} \quad C \leq 3\|x\| \quad \Longrightarrow \quad 3\|\nabla U(x)\| \leq \|\nabla U(y)\|;$$

(viii) for every $A \in \mathbb{R}$ *there exists* $R \in \mathbb{R}$ *such that, for all* $x, y \in X$,

$$R < \|x\| \quad \text{and} \quad 2^L\|x\| \leq \|y\| \quad \Longrightarrow \quad U(x) - U(y) - \frac{1}{2}\|x\|^2 < -A;$$

(ix) $L > 0$;

(x) $\varepsilon > 0$.

Then P is not geometrically ergodic with respect to π.

Proof. We first put the three growth assumptions into slightly more convenient forms. From Condition (vi), increasing the radius if necessary, for every $A \in \mathbb{R}$ there is R such that

$$R \le \|x\| \quad \Longrightarrow \quad A \le \frac{\|\nabla U(x)\|}{\|x\|}.$$

Indeed, if the strict threshold supplied by Condition (vi) is R_0, one may take $R = R_0 + 1$. Condition (vii) may be written with the two inequalities in the reverse order: there exists C such that, for all x, y,

$$C \le 3\|x\| \quad \text{and} \quad 3\|x\| \le \|y\| \quad \Longrightarrow \quad 3\|\nabla U(x)\| \le \|\nabla U(y)\|.$$

Finally, from Condition (viii), for every $B \in \mathbb{R}$ there is R such that

$$R \le \|x\| \quad \text{and} \quad 2^L\|x\| \le \|y\| \quad \Longrightarrow \quad U(x) - U(y) - \frac{1}{2}\|x\|^2 \le B :$$

apply Condition (viii) with $A = -B$, and enlarge the resulting radius by 1.

By Lemma 1, applied with $\delta = 1/2$, with Conditions (ii), (iii), (vi), (vii), (viii), (ix), (x), and with the non-strict forms just recorded, we obtain the following uniform estimate. For every $\eta > 0$ there is R_1 such that, whenever $R_1 \le \|x\|$ and $\|p\| \le \sqrt{\|x\|}$, the HMC acceptance probability at (x, p) is at most η.

The acceptance probability appearing in the kernel is exactly α. Indeed, the position and momentum after the leapfrog step are precisely the two components of $\Phi_{U,L,\varepsilon}(x, p)$. Substituting them into the Metropolis ratio and using the elementary identity $e^{a-b} = e^a e^{-b}$ gives

$$\alpha(x, p) = \min\left\{1, \exp\left(U(x) - U(q_1) + \frac{1}{2}\|p\|^2 - \frac{1}{2}\|q_2\|^2\right)\right\}.$$

Therefore the preceding estimate becomes: for every $\eta > 0$ there is R_1 such that

$$R_1 < \|x\|, \qquad \|p\| \le \sqrt{\|x\|} \quad \Longrightarrow \quad \alpha(x, p) \le \eta.$$

Also, directly from the definition of α,

$$0 \le \alpha(x, p) \le 1 \qquad \text{for all } x, p \in X.$$

By Lemma 6, for each fixed x, the function $p \mapsto \alpha(x, p)$ is measurable with respect to the

Gaussian momentum law μ_G, and each transition measure P_x is a probability measure.

Suppose, for contradiction, that P is geometrically ergodic with respect to π. Then there exist ρ with $0 \le \rho < 1$ and a Borel function $M : X \to [0, \infty]$, finite π-almost everywhere, such that

$$\|P_x^n - \pi\|_{\mathrm{TV}} \le M(x)\rho^n \qquad \text{for all } n \ge 0 \text{ and all } x.$$

Choose $\gamma \in (0, 1)$ with $\rho < 1 - \gamma$, for instance $\gamma = (1 - \rho)/2$.

Define the rejection probability at x by

$$q(x) = 1 - \int_X \alpha(x, p)\, \mu_G(dp).$$

Since μ_G is a probability measure and $0 \le \alpha(x, p) \le 1$, for each fixed x the function $\alpha(x, \cdot)$ is integrable, its integral lies in $[0, 1]$, and hence

$$0 \le q(x) \le 1 \qquad \text{for all } x.$$

By Lemma 8, there is a radius R such that

$$R < \|x\| \quad \Longrightarrow \quad 1 - \gamma \le q(x).$$

Next, π has no atoms. Indeed, by Condition (v),

$$\pi(\{x\}) = c \int_{\{x\}} e^{-U(y)}\, dy = 0,$$

because Lebesgue measure on $\mathbb{R}^d$ gives zero mass to singletons. Hence

$$\pi(\{x\}) = 0 \qquad \text{for every } x \in X.$$

By the one-step decomposition of the HMC kernel, Lemma 3, the rejected branch gives the domination

$$P_x \ge q(x)\delta_x$$

as positive measures. Explicitly,

$$P_x = \int_X \left[\alpha(x, p)\delta_{q_1(x,p)} + \big(1 - \alpha(x, p)\big)\delta_x\right] \mu_G(dp) \ge \int_X \big(1 - \alpha(x, p)\big)\delta_x\, \mu_G(dp) = q(x)\delta_x,$$

since the accepted part is nonnegative and μ_G has total mass one. Applying the iterated-

kernel domination Lemma 7 gives

$$P_x^n \geq q(x)^n \delta_x \qquad \text{for all } n \geq 0 \text{ and all } x.$$

Because π is atomless, Lemma 10 yields the total variation lower bound

$$\|P_x^n - \pi\|_{\mathrm{TV}} \geq q(x)^n \qquad \text{for all } n \geq 0 \text{ and all } x.$$

Indeed, testing against the indicator of the singleton $\{x\}$, whose π-integral is zero, the P_x^n-integral is at least $q(x)^n$.

We also need a point with large norm at which M is finite. By Condition (v) and $d > 0$, for the radius R above,

$$\pi\big(\{x : R < \|x\|\}\big) > 0.$$

This is Lemma 4: the exterior of a ball has positive Lebesgue measure, and the density ce^{-U} is strictly positive. Let

$$F = \{x : M(x) < \infty\}.$$

Since M is finite π-almost everywhere, $\pi(F^c) = 0$. Hence removing the π-null set F^c does not change the measure:

$$\pi\big(\{x : R < \|x\|\} \cap F\big) = \pi\big(\{x : R < \|x\|\}\big) > 0.$$

In particular this intersection is nonempty, so choose $x \in X$ with

$$R < \|x\|, \qquad M(x) < \infty.$$

For this x, the rejection estimate gives

$$1 - \gamma \leq q(x).$$

Combining the lower bound for the total variation distance with the assumed geometric bound, for every $n \geq 0$,

$$(1-\gamma)^n \leq q(x)^n \leq \|P_x^n - \pi\|_{\mathrm{TV}} \leq M(x)\rho^n,$$

where the last inequality is meaningful as an ordinary real inequality because $M(x) < \infty$.

If $\rho = 0$, then taking $n = 1$ gives

$$1 - \gamma \le \|P_x - \pi\|_{\mathrm{TV}} \le M(x) \cdot 0 = 0,$$

which is impossible because $\gamma < 1$.

If $\rho > 0$, set

$$c_0 = \frac{1 - \gamma}{\rho}.$$

Since $\rho < 1 - \gamma$, we have $c_0 > 1$. Dividing the inequality

$$(1 - \gamma)^n \le M(x)\rho^n$$

by the positive number ρ^n, we obtain

$$c_0^n = \left(\frac{1 - \gamma}{\rho}\right)^n \le M(x) \qquad \text{for all } n \ge 0.$$

But $c_0 > 1$, so $c_0^n \to \infty$, while $M(x) < \infty$ is fixed. Choosing n with $M(x) < c_0^n$ gives a contradiction.

Thus the assumed geometric ergodicity is impossible. Therefore P is not geometrically ergodic with respect to π.

□

Task outcomes and dependencies

Task registry snapshot: paper_fresh_20261028t151409527z_hmc_v5, revision 141, captured 2026-10-28T18:08:14.500Z.

This report belongs to paper_fresh_20261028t151409527z_hmc_v5 (job paper_fresh_20261028t15140 its own terminal report determines its outcome, since registry

settlement can lag reporting.

paper_53e951f48c854f4db91e6ecda697ee2b (job job_paper_53e951f48c854f4db91e6ecda697ee2b): proved; proof-ready; dependencies: paper_a62240be95074e8caf07c24daf22159a.

paper_4f562633bddc4ee9afee32ec40dfa0f0 (job job_paper_4f562633bddc4ee9afee32ec40dfa0f0): proved; proof-ready; dependencies: none.

paper_a9a2ab5d4e094509b2cdb371b3b02f1c (job job_paper_a9a2ab5d4e094509b2cdb371b3b02f1c): proved; proof-ready; dependencies: none.

paper_7ad7b898530c48b2be48d9fe88a3d538 (job job_paper_7ad7b898530c48b2be48d9fe88a3d538): proved; proof-ready; dependencies: none.

paper_a62240be95074e8caf07c24daf22159a (job job_paper_a62240be95074e8caf07c24daf22159a): proved; proof-ready; dependencies: paper_b7909512ace54739b0ef8eae60aa82ed.

paper_6ac1720d15ae46eb9cc770b0a5541a93: external; pending; dependencies: none.

paper_b7909512ace54739b0ef8eae60aa82ed (job job_paper_b7909512ace54739b0ef8eae60aa82ed): proved; proof-ready; dependencies: none.

Lemma proved by task paper_53e951f48c854f4db91e6ecda697ee2b (job job_paper_53e951f48c854f4db9 interface revision 1. See that task's own proof

and summary. Its remaining dependencies still apply; the consumer's sealed axiom interface is unchanged.

Each task has its own outcome. A proved child does not certify the main task or discharge other external assumptions.